Slip 1

Q.1) Write an application to create a splash screen.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:background="#2196F3"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Welcome to Mobile App"
      android:textSize="28sp"
      android:textStyle="bold"
      android:textColor="#FFFFFF"/>
</RelativeLayout>
```

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splash_screen.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:background="#000000">

   <ImageView
      android:id="@+id/imageView"
      android:layout_width="150dp"
      android:layout_height="150dp"
      android:src="@mipmap/ic_launcher"
      android:layout_centerInParent="true"/>

   <TextView
      android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:text="Splash Screen"
    android:textSize="24sp"
    android:textColor="#FFFFFF"
    android:layout_below="@id/imageView"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="20dp"/>
</RelativeLayout>
```

SplashActivity.java

```
package com.example.splashscreen;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;

import androidx.appcompat.app.AppCompatActivity;

public class SplashActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.splash_screen);

    new Handler().postDelayed(new Runnable() {
      @Override
      public void run() {

        Intent intent = new Intent(SplashActivity.this,
            MainActivity.class);

        startActivity(intent);
        finish();
      }
    },3000);
  }
}
```

MainActivity.java

```java
package com.example.splashscreen;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

AndroidManifest.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.splashscreen">

    <application
        android:allowBackup="true"
        android:label="SplashScreen"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity android:name=".MainActivity"/>

        <activity android:name=".SplashActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

    </application>

</manifest>
```

Q.2) Create table Student (roll_no, name, address, percentage). Create Application for performing the following operation on the table. (Using SQLite database). i] Insert record of 5 new student details. ii]Show all the student details.

DBHelper.java

```
package com.example.studentdb;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

  public static final String DBNAME = "StudentDB";

  public DBHelper(Context context) {
    super(context, DBNAME, null, 1);
  }

  @Override
  public void onCreate(SQLiteDatabase db) {

    db.execSQL("CREATE TABLE Student(" +
        "roll_no INTEGER PRIMARY KEY," +
        "name TEXT," +
        "address TEXT," +
        "percentage REAL)");
  }

  @Override
  public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

    db.execSQL("DROP TABLE IF EXISTS Student");
    onCreate(db);
  }

  // Insert Record
  public Boolean insertData(int roll, String name,
                String address, float percentage) {
```

```java
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues cv = new ContentValues();

        cv.put("roll_no", roll);
        cv.put("name", name);
        cv.put("address", address);
        cv.put("percentage", percentage);

        long result = db.insert("Student", null, cv);

        if(result == -1)
            return false;
        else
            return true;
    }

    // Show All Records
    public Cursor getData() {

        SQLiteDatabase db = this.getWritableDatabase();

        Cursor cursor = db.rawQuery(
                "SELECT * FROM Student", null);

        return cursor;
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">
```

```xml
    <EditText
      android:id="@+id/roll"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Roll No"/>

    <EditText
      android:id="@+id/name"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"/>

    <EditText
      android:id="@+id/address"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Address"/>

    <EditText
      android:id="@+id/percentage"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Percentage"/>

    <Button
      android:id="@+id/insertBtn"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Insert"/>

    <Button
      android:id="@+id/viewBtn"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="View All"/>
  </LinearLayout>

</ScrollView>
```

MainActivity.java

```java
package com.example.studentdb;

import androidx.appcompat.app.AppCompatActivity;

import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

  EditText roll, name, address, percentage;
  Button insertBtn, viewBtn;

  DBHelper DB;

  @Override
  protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    roll = findViewById(R.id.roll);
    name = findViewById(R.id.name);
    address = findViewById(R.id.address);
    percentage = findViewById(R.id.percentage);

    insertBtn = findViewById(R.id.insertBtn);
    viewBtn = findViewById(R.id.viewBtn);

    DB = new DBHelper(this);

    // Insert 5 Records
    insertBtn.setOnClickListener(new View.OnClickListener() {
      @Override
      public void onClick(View v) {

        Boolean checkInsert = DB.insertData(
            Integer.parseInt(roll.getText().toString()),
            name.getText().toString(),
```

```
            address.getText().toString(),
            Float.parseFloat(percentage.getText().toString())
        );

        if(checkInsert == true)
            Toast.makeText(MainActivity.this,
                    "Data Inserted", Toast.LENGTH_SHORT).show();
        else
            Toast.makeText(MainActivity.this,
                    "Insertion Failed", Toast.LENGTH_SHORT).show();
    }
});

// Show All Records
viewBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        Cursor res = DB.getData();

        if(res.getCount() == 0) {
            Toast.makeText(MainActivity.this,
                    "No Data Found", Toast.LENGTH_SHORT).show();
            return;
        }

        StringBuffer buffer = new StringBuffer();

        while(res.moveToNext()) {

            buffer.append("Roll No : "
                    + res.getString(0) + "\n");

            buffer.append("Name : "
                    + res.getString(1) + "\n");

            buffer.append("Address : "
                    + res.getString(2) + "\n");

            buffer.append("Percentage : "
                    + res.getString(3) + "\n\n");
        }
```

```
            Toast.makeText(MainActivity.this,
                buffer.toString(),
                Toast.LENGTH_LONG).show();
        }
      });
  }
}
```

Sample Student Records

Roll No	Name	Address	Percentage
1	Amit	Pune	85.5
2	Rahul	Mumbai	78.0
3	Sneha	Nashik	91.2
4	Priya	Nagpur	88.4
5	Karan	Solapur	75.6